ECE 329
Lecture 18

Uniform Plane Waves in Free Space

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Summary of Maxwell's Equations

Faraday's Law $\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$ $\quad \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$

Ampere's Law $\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$ $\quad \nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$

Gauss' Law $\oiint_S \vec{B} \bullet d\vec{S} = 0$ $\quad \nabla \bullet \vec{B} = 0$

Gauss' Law $\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$ $\quad \nabla \bullet \vec{D} = \rho$

Continuity Eq. $\oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$ $\quad \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$

Motivation for Waves

- Maxwell's equations say…
  - Time variation in **J**(t) leads to spatial variation of **H**(t)

$$\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

  - Time variation in **H**(t) leads to spatial variation in **E**(t)

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

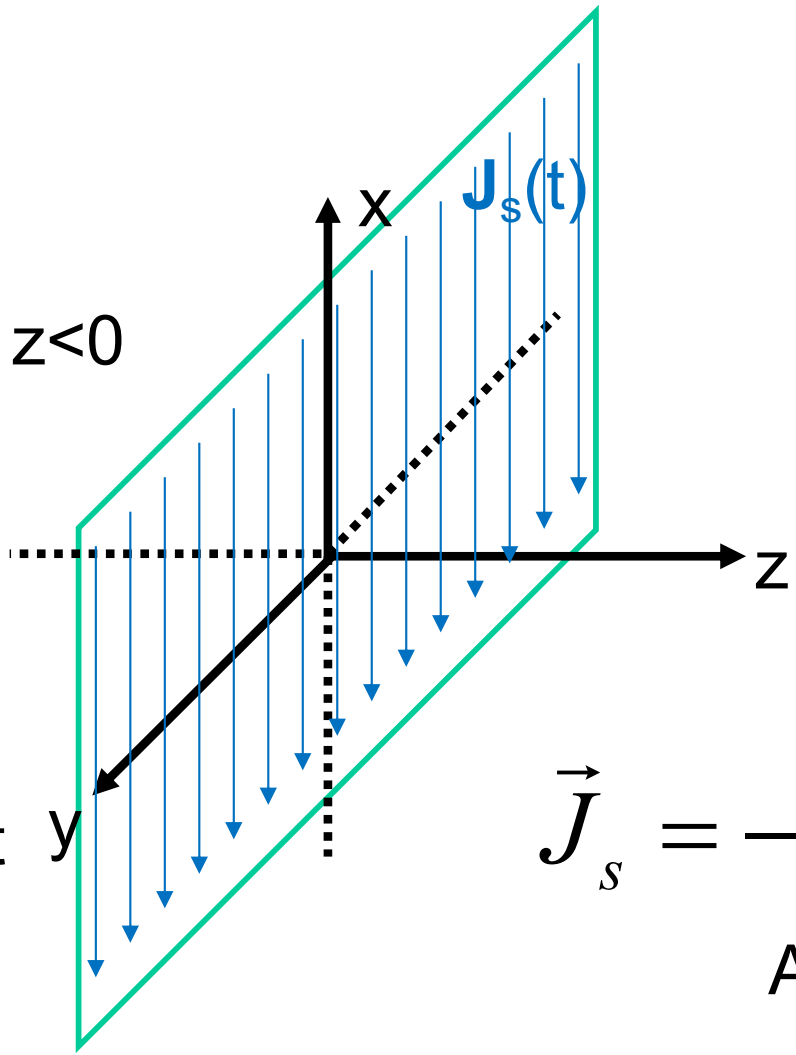
  - Time variation in **E**(t) leads to spatial variation in **H**(t)

$$\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

- This recursion relationship between **E**(t) and **H**(t) leads to electromagnetic wave propagation

Field Source

- Infinite plane sheet of current at z=0



Free space z<0

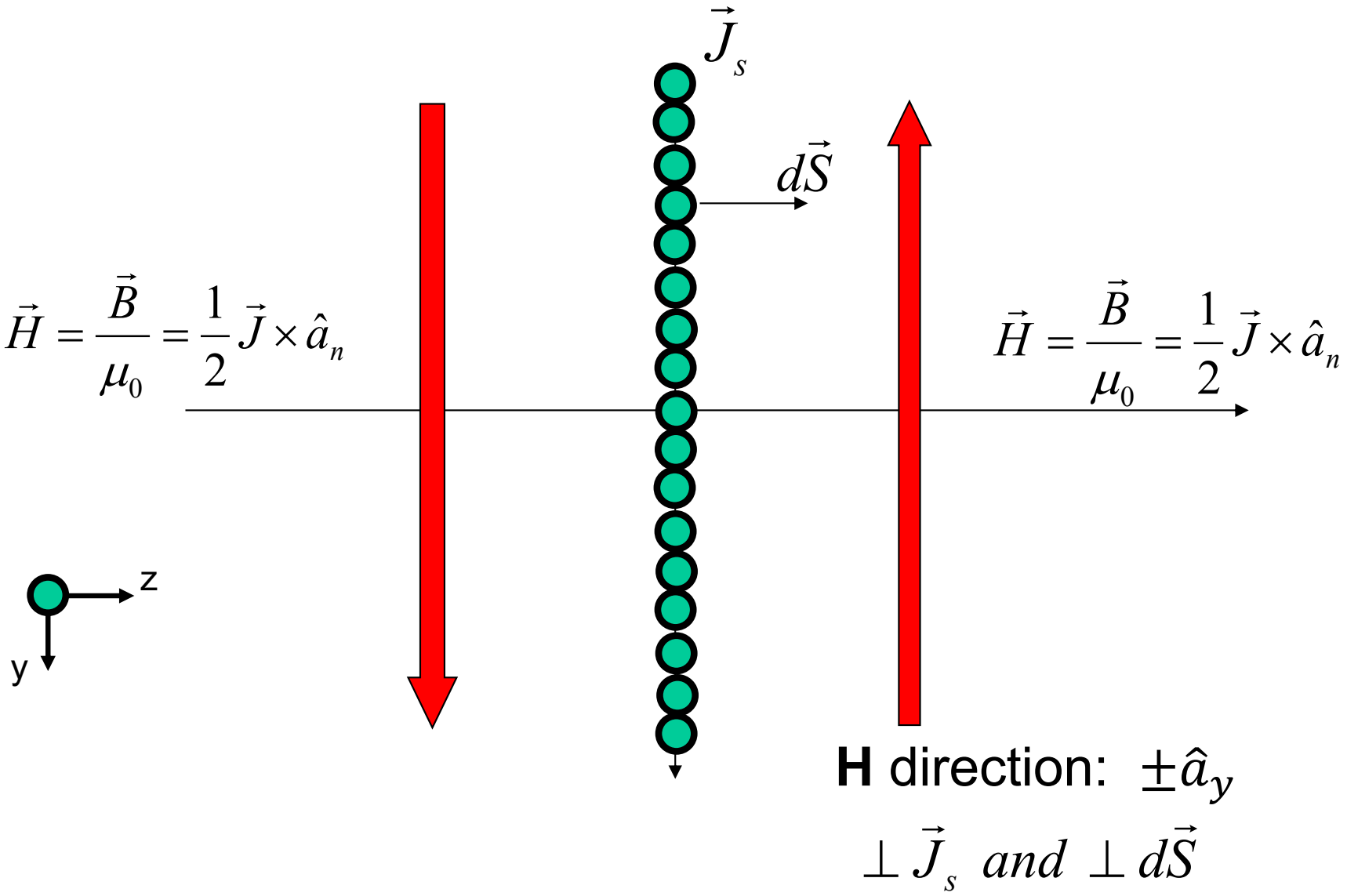
Free space z>0

J_s = surface current in Amps per meter

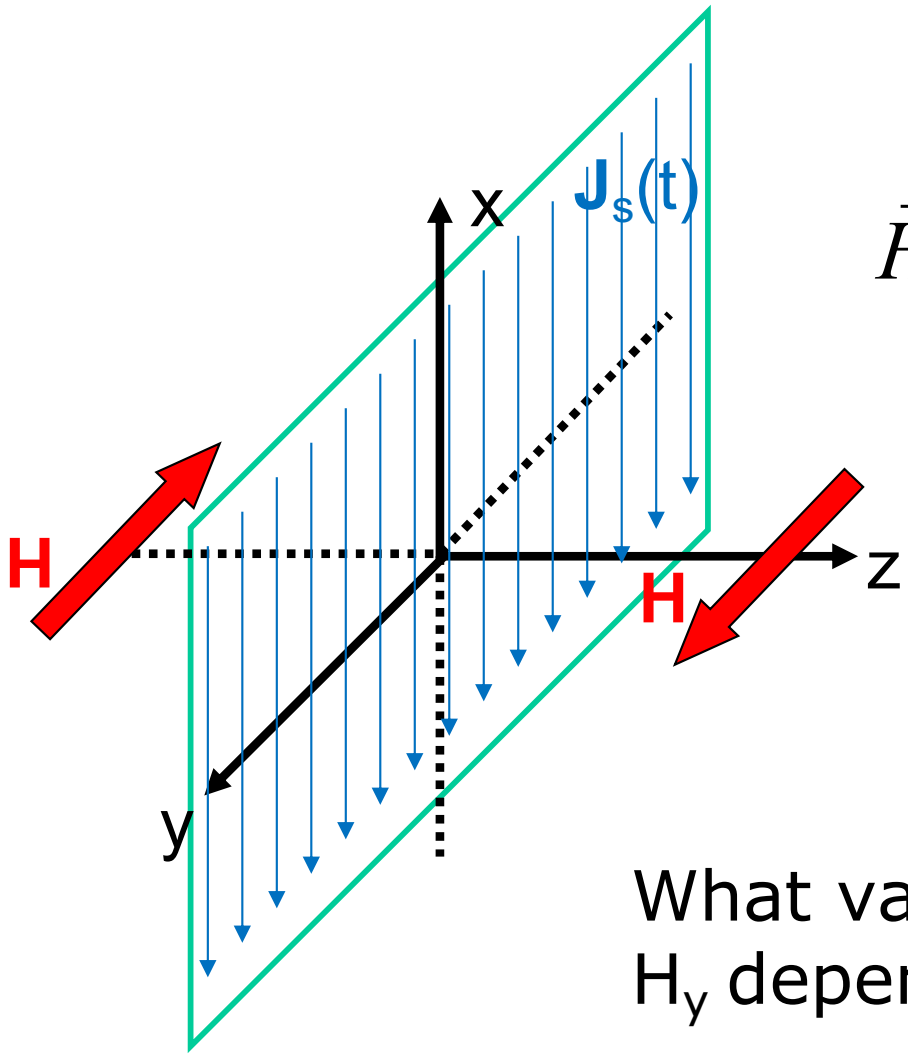
$$\vec{J}_s = -J_s(t)\hat{a}_x$$

At z=0

Hmm, we did the static case



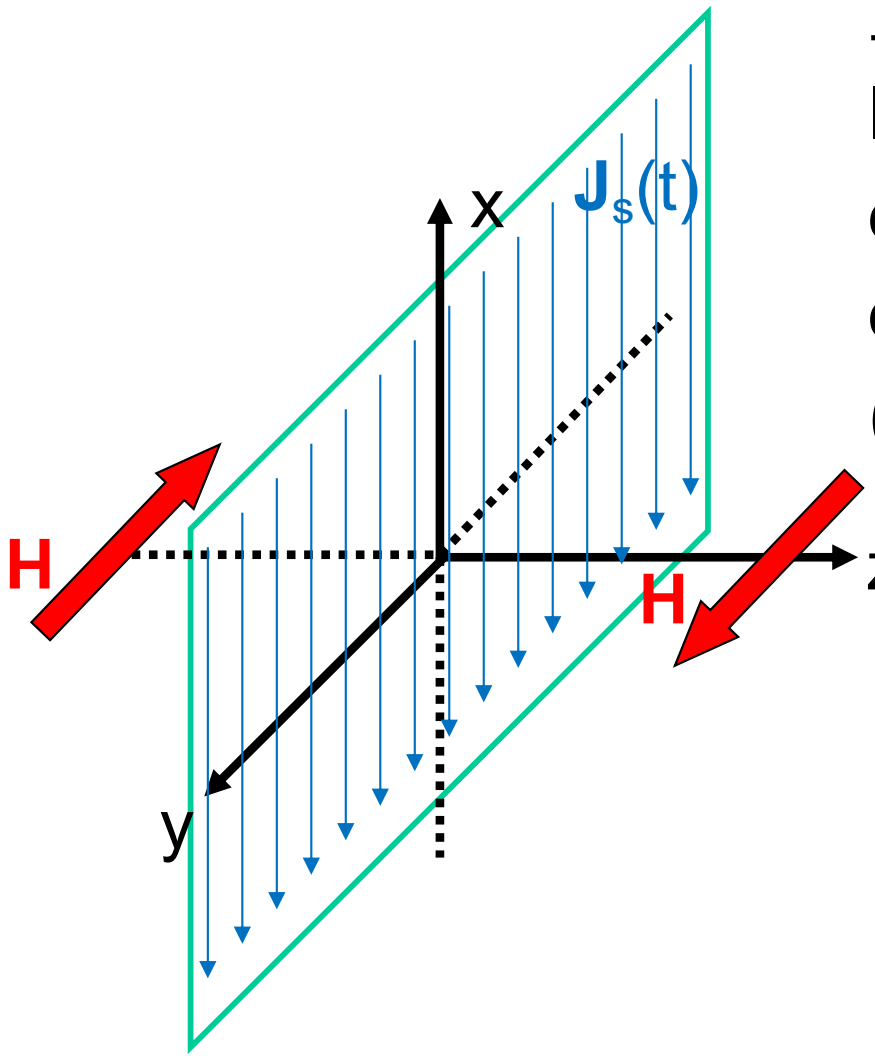
So we expect **H** to be along the ±y-direction



$$\vec{H} = H_y \hat{a}_y$$

What variables can H_y depend on??

H_y only depends on z and t



Infinite plane looks the same everywhere so H_y can't depend on (x,y) coordinates

$$\vec{H} = H_y(z,t)\hat{a}_y$$

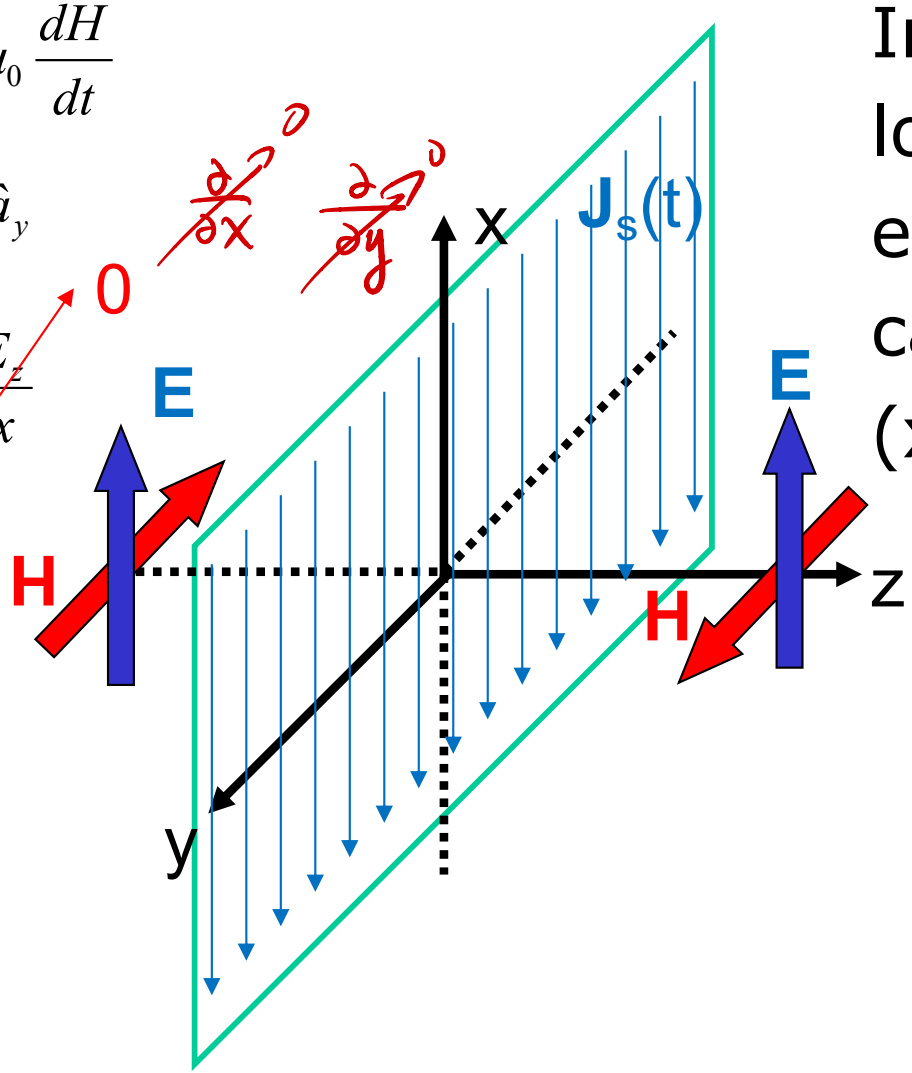
E depends only on z and t, but in what direction is **E**?

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt} = -\mu_0 \frac{d\vec{H}}{dt}$$

$$= -\mu_0 \frac{dH_y}{dt} \hat{a}_y$$

$$\left(\nabla \times \vec{E}\right)_y = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}$$

$$\frac{\partial E_x}{\partial z} = -\mu_0 \frac{dH_y}{dt}$$



Infinite plane looks the same everywhere so **E** can't depend on (x,y) coordinates

$E_x \neq 0$, what about E_y and E_z?

$$0 = -\frac{dB_x}{dt} = \left(\nabla \times \vec{E}\right)_x = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}$$

$$So\ \frac{\partial E_y}{\partial z} = 0 \Rightarrow E_y = const,$$

$but\ const = 0$ since we can turn off our source, i.e. set J(t) = 0.

$$0 = -\frac{dB_z}{dt} = \left(\nabla \times \vec{E}\right)_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}$$

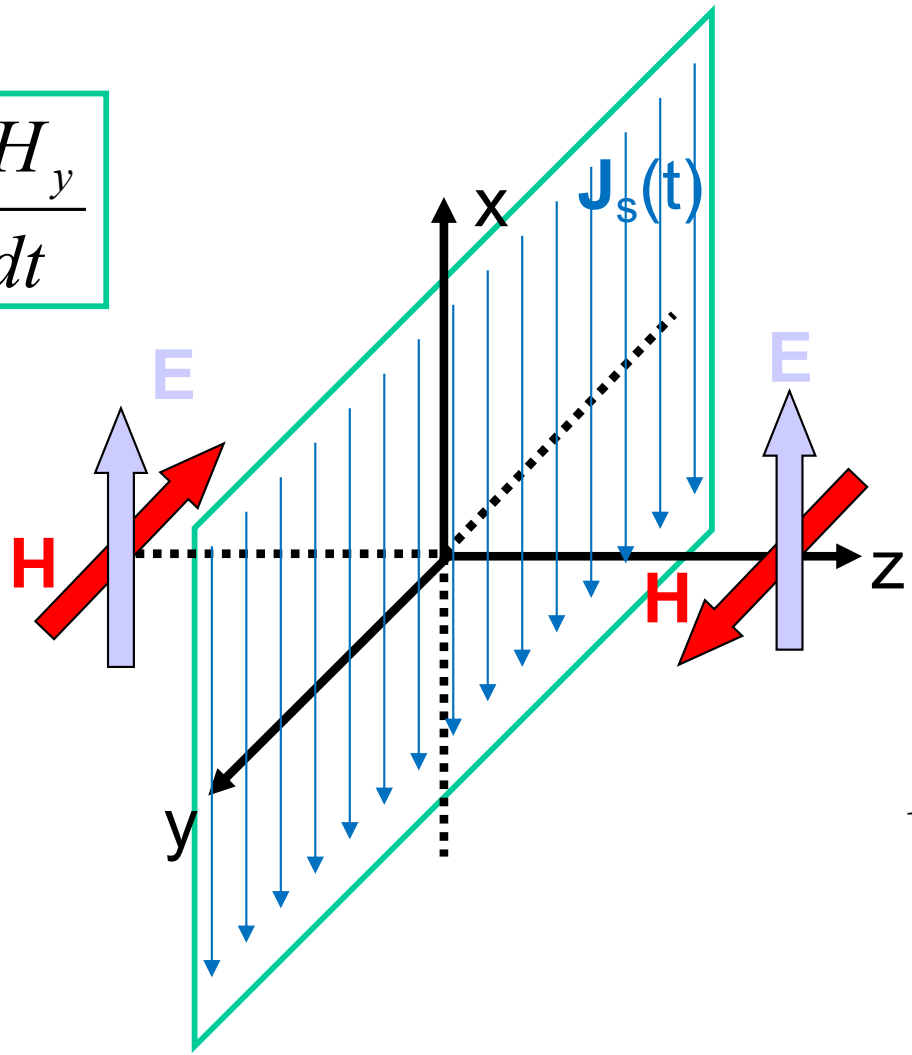
No new information.

$$0 = \frac{\rho}{\varepsilon_0} = \nabla \bullet \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$So\ \frac{\partial E_z}{\partial z} = 0 \Rightarrow E_z = const = 0$$

E then must be along the x-direction

$$\frac{\partial E_x}{\partial z} = -\mu_0 \frac{dH_y}{dt}$$



$\vec{E}$ always opposite of $\vec{J}_s$

$$\vec{E} = E_x(z,t)\hat{a}_x$$

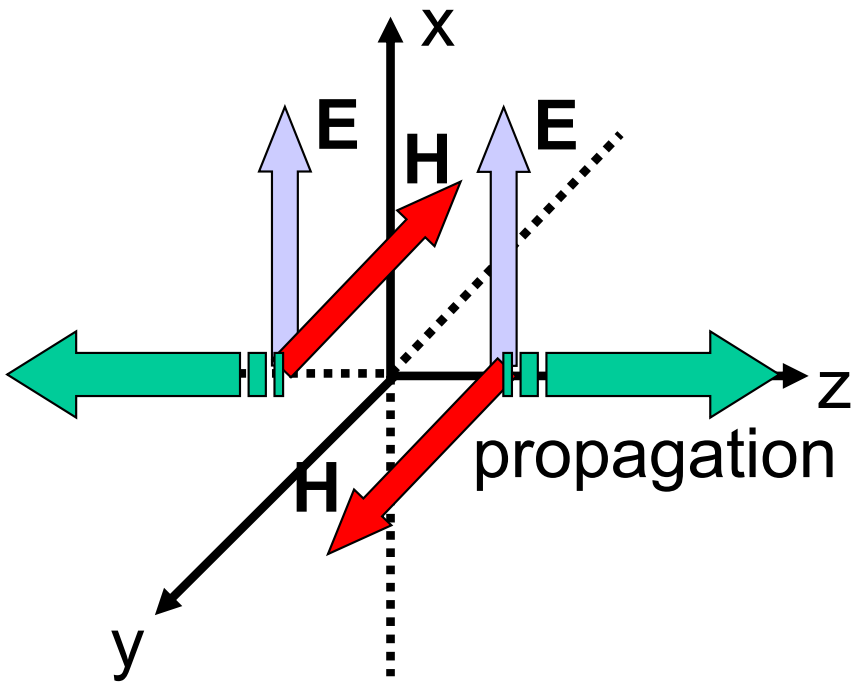
Summary so far …

- We have **E** and **H** that are
  - Perpendicular to each other
  - Perpendicular to the direction of propagation $\vec{E}\times\vec{H}\rightarrow$ propagation direction.
  - Magnitude is constant ("uniform") in any plane perpendicular to the propagation direction

$$\vec{J}_S = -J_x(t)\hat{a}_x$$

$$\vec{E} = E_x(z,t)\hat{a}_x$$

$$\vec{H} = H_y(z,t)\hat{a}_y$$



Following Maxwell's Footsteps

- We will SIMULTANEOUSLY solve Maxwell's equations to find **E** and **H** caused by **J**

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{E} = E_x(z,t)\hat{a}_x$$

$$\vec{H} = H_y(z,t)\hat{a}_y$$

$$\vec{J}_s = -J_x(t)\hat{a}_x$$

Apply the two Maxwell's Equations

- Performing the cross product, only two equations contain E_x or H_y

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t}$$

$$\frac{\partial H_y}{\partial z} = -J_x - \frac{\partial D_x}{\partial t}$$

↑ in z=0 plane.

Simplify to **E** and **H**

- Use constitutive relationships $\mathbf{D}=\varepsilon_0\mathbf{E}$ and $\mathbf{B}=\mu_0\mathbf{H}$ for free space to express **D** and **B** in terms of **E** and **H**
- We write the equations for everywhere EXCEPT at z=0 (where the current source is) so J_x goes away - for now

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t} = -\mu_0 \frac{\partial H_y}{\partial t}$$

$$\frac{\partial H_y}{\partial z} = -\frac{\partial D_x}{\partial t} = -\varepsilon_0 \frac{\partial E_x}{\partial t}$$

$z \gtrless 0$

Eliminate **H** in favor of **E** only

$$\frac{\partial E_x}{\partial z} = -\mu_0 \frac{\partial H_y}{\partial t}$$

$$\frac{\partial H_y}{\partial z} = -\varepsilon_0 \frac{\partial E_x}{\partial t}$$

$$\frac{\partial^2 E_x}{\partial z^2} = -\mu_0 \frac{\partial}{\partial z}\left(\frac{\partial H_y}{\partial t}\right) = -\mu_0 \frac{\partial}{\partial t}\left(\frac{\partial H_y}{\partial z}\right) = -\mu_0 \frac{\partial}{\partial t}\left(-\varepsilon_0 \frac{\partial E_x}{\partial t}\right)$$

- Coupled 1st order partial differential equation (PDE) → single 2nd order PDE

This is the "wave equation"

$$\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_x}{\partial t^2}$$

- Why is it called a wave equation?
  - The solution to this differential equation will be a function whose shape moves like a wave in the z-direction
  - How do we solve it?
    - Two techniques: Separation of variables (see book) or factorable operators (my notes)
  - We will do the solution for **E**, and then we can come back and plug the solution into Maxwell's equation to get **H**

Change of variables

$$\frac{1}{\mu_0\varepsilon_0}\frac{\partial^2 E_x}{\partial z^2} = \mu_0\varepsilon_0\frac{\partial^2 E_x}{\partial t^2}$$

- Makes the math a little cleaner

$$\tau = z\sqrt{\mu_0\varepsilon_0}$$

Has units of time

$$\frac{\partial^2 E_x}{\partial \tau^2} = \frac{\partial^2 E_x}{\partial t^2}$$

Still the wave eqn

Two possible solutions

$$\frac{\partial^2 E_x}{\partial \tau^2} = \frac{\partial^2 E_x}{\partial t^2}$$

$$\frac{\partial^2 E_x}{\partial \tau^2} - \frac{\partial^2 E_x}{\partial t^2} = 0$$

$$"x^2 - y^2 = (x+y)(x-y)"$$

$$\left(\frac{\partial}{\partial \tau} + \frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial \tau} - \frac{\partial}{\partial t}\right)E_x = 0$$

$$\frac{\partial E_x}{\partial \tau} = -\frac{\partial E_x}{\partial t} \qquad \frac{\partial E_x}{\partial \tau} = +\frac{\partial E_x}{\partial t}$$

Now we have two first order diff eqns that each can be a solution to the wave equation

Two possible solutions

$Af' \cdot (-1) = -Af'$

$$\frac{\partial E_x}{\partial \tau} = -\frac{\partial E_x}{\partial t}$$

If E_x is a function of $(t-\tau)$ the equation is satisfied

$$E_x(\tau, t) = Af(t-\tau)$$

A = a constant

$Bg' = Bg'$

$$\frac{\partial E_x}{\partial \tau} = +\frac{\partial E_x}{\partial t}$$

If E_x is a function of $(t+\tau)$ the equation is satisfied

$$E_x(\tau, t) = Bg(t+\tau)$$

B = a constant

Combining the two possible solutions…

$$E_x(\tau, t) = Af(t-\tau) + Bg(t+\tau)$$

Changing variables *back* to *z*

$$E_x(\tau,t) = Af(t-\tau) + Bg(t+\tau)$$

$$E_x(z,t) = \underbrace{Af\left(t - z\sqrt{\mu_0\varepsilon_0}\right)}_{\text{Traveling wave propagating in the +z direction}} + \underbrace{Bg\left(t + z\sqrt{\mu_0\varepsilon_0}\right)}_{\text{Traveling wave propagating in the -z direction}}$$

Solving for **H**

$$E_x(z,t) = Af(t - z\sqrt{\mu_0\varepsilon_0}) + Bg(t + z\sqrt{\mu_0\varepsilon_0})$$

Recall…

$$\frac{\partial H_y}{\partial z} = -\varepsilon_0 \frac{\partial E_x}{\partial t}$$

$$H_y(z,t) = \frac{1}{\sqrt{\mu_0/\varepsilon_0}}\left[Af(t - z\sqrt{\mu_0\varepsilon_0}) - Bg(t + z\sqrt{\mu_0\varepsilon_0})\right]$$

$$\frac{E^+}{H^+} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \quad \Omega \qquad = -\frac{E^-}{H^-}$$

Two definitions

$$v_p = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \text{speed of light} = 3 \times 10^8 \ (m/s) = 300 \ m/\mu s$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377 \ (ohms) \approx 120\pi \ \Omega$$

Intrinsic impedance of free space

material $v_p = \frac{1}{\sqrt{\mu\varepsilon}}$

$\eta = \sqrt{\frac{\mu}{\varepsilon}}$

Rewriting the solution

$$E_x(z,t) = Af\left(t - \frac{z}{v_p}\right) + Bg\left(t + \frac{z}{v_p}\right)$$

$$H_y(z,t) = \frac{1}{\eta_0}\left[Af\left(t - \frac{z}{v_p}\right) - Bg\left(t + \frac{z}{v_p}\right) \right]$$

Solution is a superposition of traveling waves

- one going in +z direction
- one going in -z direction

Challenge Problem: Velocity of propagation

- The velocities of propagation for the following waves: $f = (0.05y-t)^2$, $g = u(t+0.02x)$, $h = \cos(2\pi 10^8 t - 2\pi z)$ are:

(a) $0.05\mathbf{a}_y$, $-0.02\mathbf{a}_x$, $10^8\ \mathbf{a}_z$

(b) $0.05\mathbf{a}_y$, $-0.02\mathbf{a}_x$, $-10^8\ \mathbf{a}_z$

(c) $20\mathbf{a}_y$, $-50\mathbf{a}_x$, $10^8\ \mathbf{a}_z$

(d) $20\mathbf{a}_y$, $50\mathbf{a}_x$, $10^8\ \mathbf{a}_z$

(e) $20\mathbf{a}_y$, $-50\mathbf{a}_x$, $-10^8\ \mathbf{a}_z$

$f(t \mp \frac{z}{v_p})$.

a). $f = (t - \frac{y}{20})^2$.

$+20\hat{y}$

b). $g = u(t + \frac{x}{50})$.

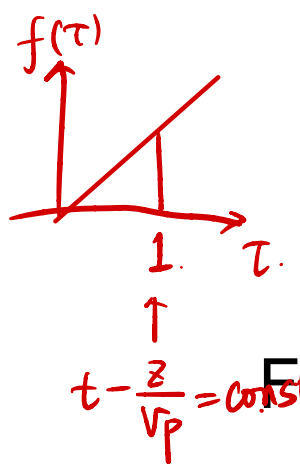
$-50\hat{x}$

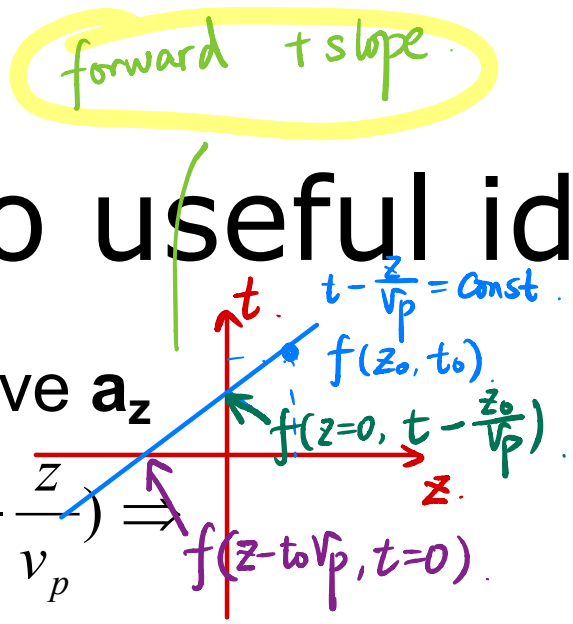
c) $h = \cos(2\pi \times 10^8 (t - \frac{z}{10^8}))$

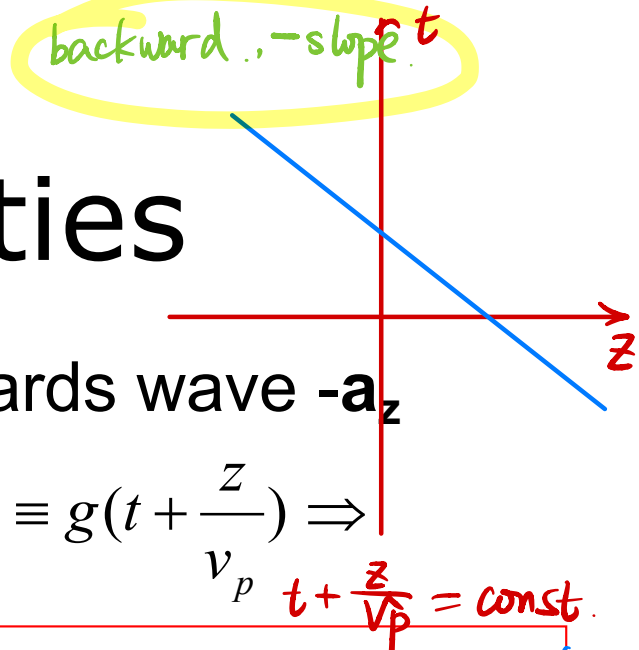
$+10^8 \hat{z}$

From D3.11 (p 171) of old book

Two useful identities







Forward wave $\mathbf{a_z}$

$$E_x(z,t) \equiv f(t-\frac{z}{v_p}) \Rightarrow$$

$$E_x(z,t) = E_x(0,t-\frac{z}{v_p}) = E_x(z-v_p t,0)$$

Backwards wave $\mathbf{-a_z}$

$$E_x(z,t) \equiv g(t+\frac{z}{v_p}) \Rightarrow$$

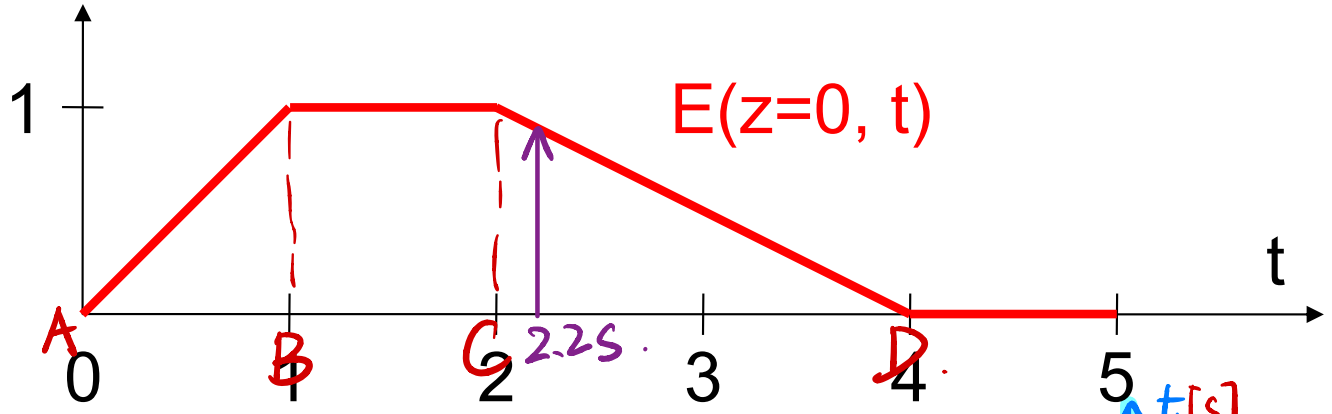
$$E_x(z,t) = E_x(0,t+\frac{z}{v_p}) = E_x(z+v_p t,0)$$

These identities simply say that the forward wave moves like $z=v_p t$ and the backwards wave like $z=-v_p t$

They allow you to express the wave solutions in terms of the wave at a fixed position or at a fixed time

Moving waveform

- A wave traveling in the $-\mathbf{a}_z$ direction with speed 100 m/s is measured at z=0:

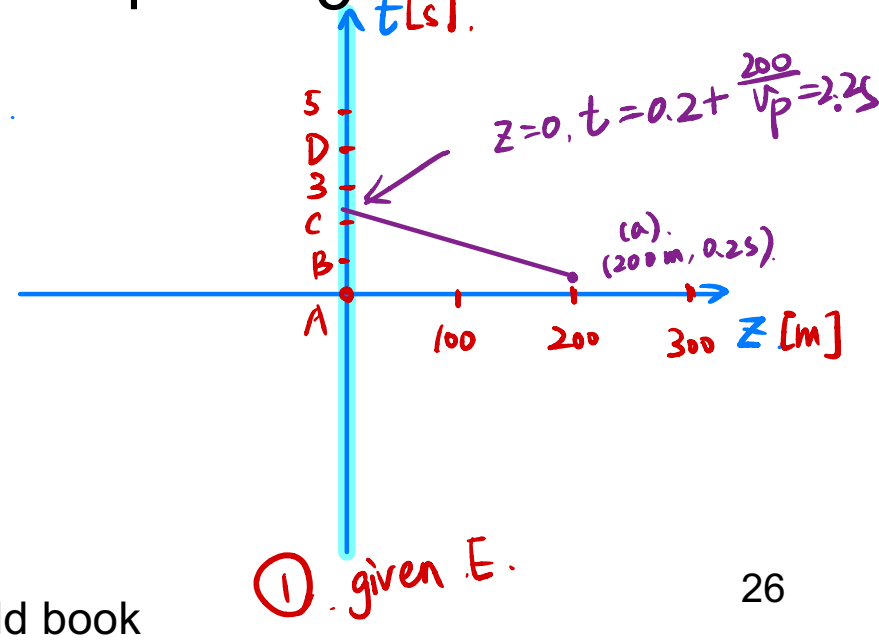


Find the wave amplitude at:

(a) z=200m, t=0.2s

(b) z=-300m, t=3.4s

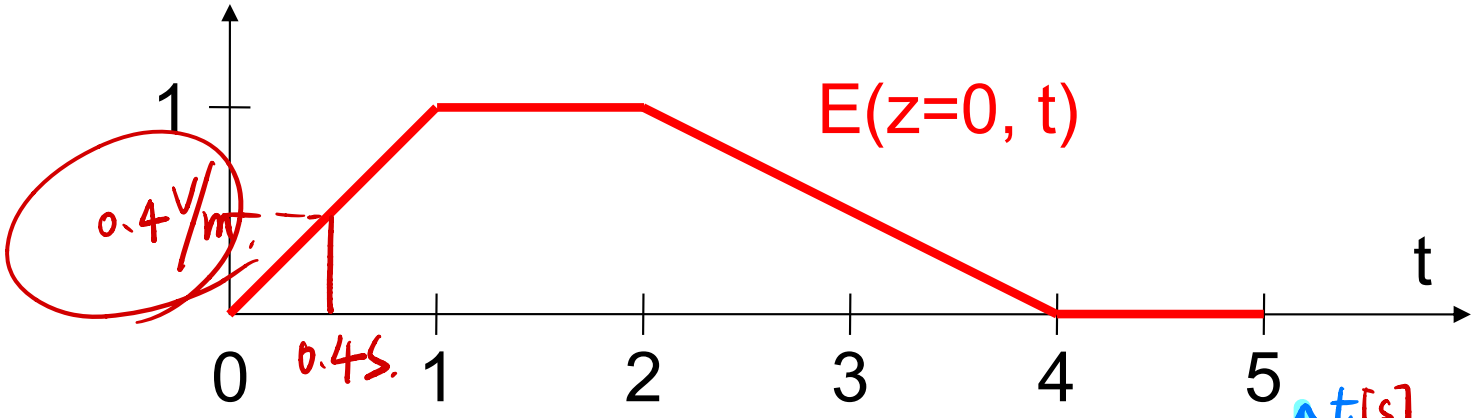
(c) z=100m, t=0.6s



From D3.13 (p 171) of old book

Moving waveform

- A wave traveling in the $-\mathbf{a}_z$ direction with speed 100 m/s is measured at z=0:

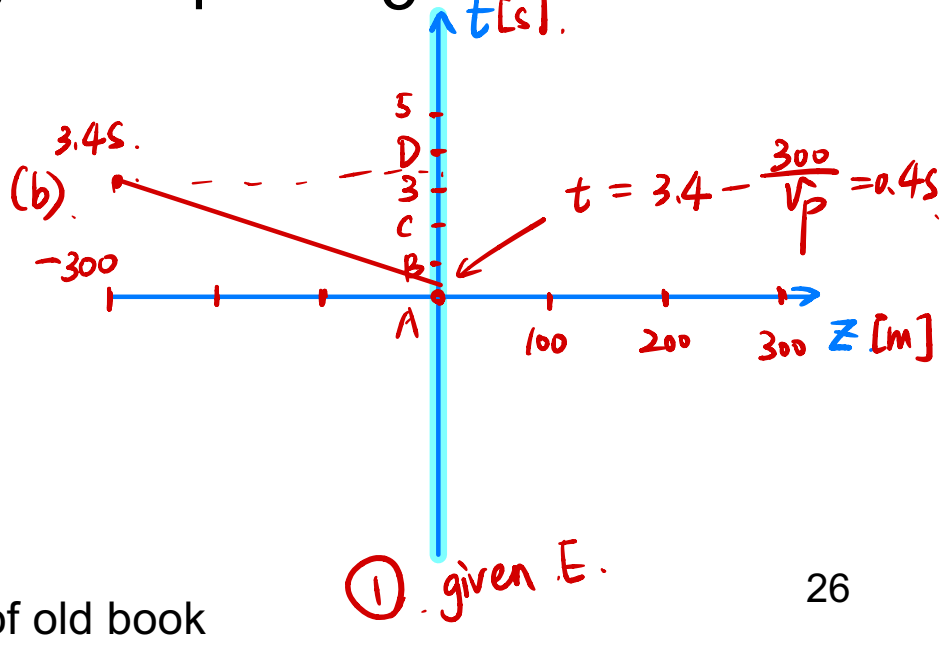


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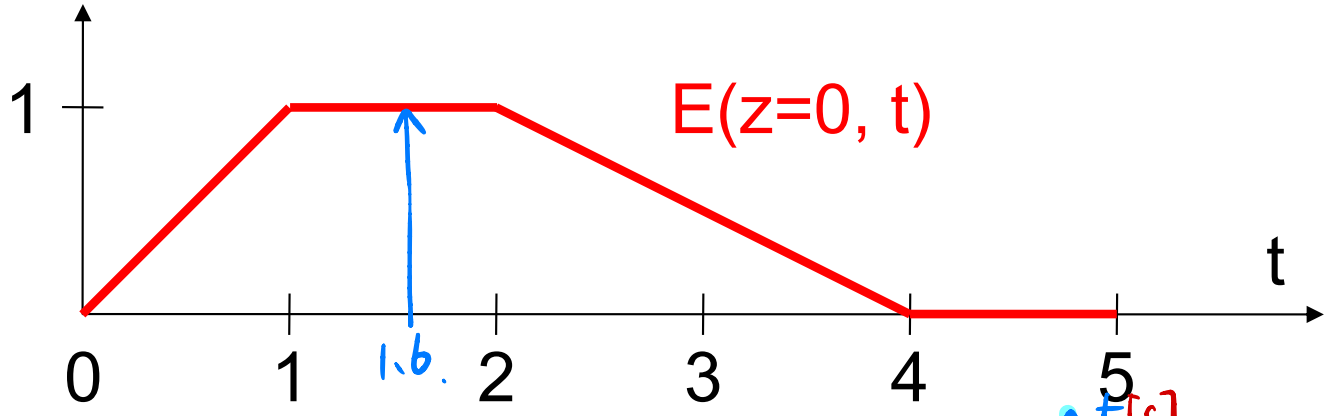
(c) z=100m, t=0.6s



From D3.13 (p 171) of old book

Moving waveform

- A wave traveling in the $-\mathbf{a}_z$ direction with speed 100 m/s is measured at z=0:

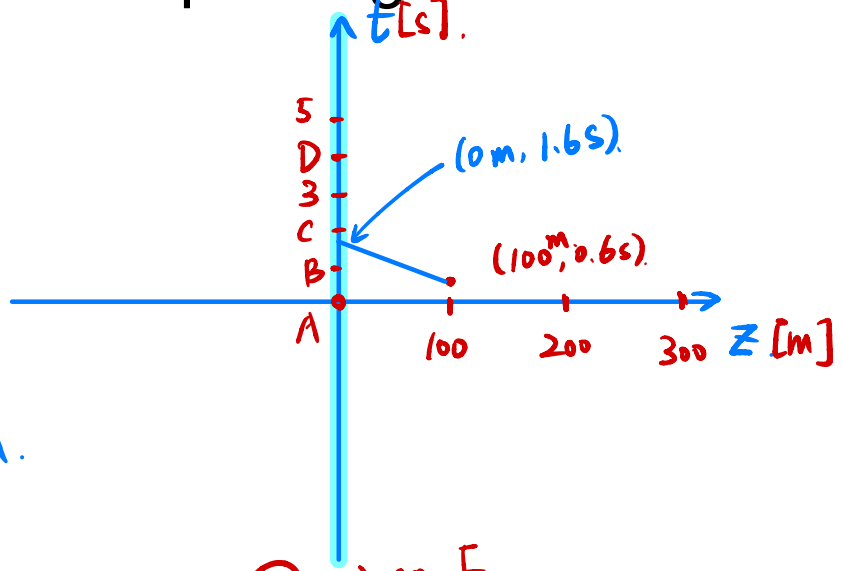


Find the wave amplitude at:

(a) z=200m, t=0.2s

(b) z=-300m, t=3.4s

(c) z=100m, t=0.6s $E \rightarrow 1$ V/m.



From D3.13 (p 171) of old book

Lecture 18 Summary

- Differentiate Maxwell's Equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

to derive wave equation: $\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_x}{\partial t^2}$

which has solution:

$$E_x(z,t) = \underbrace{Af\left(t - \frac{z}{v_p}\right)}_{\text{Traveling wave +z direction}} + \underbrace{Bg\left(t + \frac{z}{v_p}\right)}_{\text{Traveling wave -z direction}}$$